Requirements Gathering

Smart Healthcare Platform (working title)

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1. Stakeholder Analysis

1.1 Introduction

Stakeholders are the people or groups who use the system, are affected by it, or are responsible for it. This document identifies them and their main needs.

1.2 Stakeholders

Stakeholder	Description	Needs and Expectations	Influence	Interest
Patients	Main users who own a medical record	Secure and private record, full control over who can see it, easy appointment booking, finding a suitable doctor	High	High
Doctors	Medical professionals who treat patients	Access to patient history only with permission, writing diagnoses and prescriptions, clear analytics about their patients	High	High
Receptionist	Clinic staff linked to a doctor; manages appointments and the waiting queue	Easy booking, rescheduling and cancelling of appointments, a clear daily queue with patient check-in, and a way to request patient access (via OTP or QR) without seeing any medical data	Medium	High
Laboratories / Radiology Centers	Providers of tests and imaging	Simple way to upload results to the right patient, limited access to data	Medium	Medium
System Administrator	Manages the platform	Control accounts, verify doctors' licenses, monitor performance without seeing medical data	High	High
Development Team	Bimen Hanna, Bavly Hany, Aml Mahrous, Mona Mohamed	Clear requirements, realistic timeline, successful delivery	High	High
DEPI Lecturer / Evaluators	Review and grade the project	Complete documents, working system, deadlines respected	High	High

2. User Stories & Use Cases

2.1 User Stories

ID	As a…	I want to…	So that…
US-01	Patient	register and log in securely	my medical data is protected
US-02	Patient	view my full medical record (diagnoses, prescriptions, lab results)	I can follow my health in one place
US-03	Patient	search for doctors by specialty and location on a map	I can find a suitable doctor near me
US-04	Patient	book, view, and cancel appointments	I can manage my visits easily
US-05	Patient	grant or revoke a doctor's access to my record	I control who sees my data
US-06	Patient	enter my symptoms and get a suggested specialty with available doctors	I know which doctor to book
US-07	Doctor	see the list of patients who gave me access	I can open their records
US-08	Doctor	view a permitted patient's medical history	I can make a better diagnosis
US-09	Doctor	add a diagnosis and an electronic prescription	the patient's record is up to date
US-10	Doctor	see charts of common diseases and patient adherence	I can understand my patients' conditions
US-11	Doctor	track a patient's condition over time (e.g., blood sugar, blood pressure)	I can see if the treatment works
US-12	Doctor	add a receptionist account to my clinic and set its permissions	my clinic staff can manage appointments without seeing medical data
US-13	Receptionist	book, reschedule, and cancel appointments for patients	the clinic schedule stays organized
US-14	Receptionist	see the daily queue and check patients in when they arrive	the doctor knows who is waiting and the visit flow runs smoothly
US-15	Receptionist	request access to a patient's record for the doctor using OTP or QR code	the patient approves the access himself and the doctor can open the file
US-16	Receptionist	see only administrative data (name, phone, appointment)	patient medical privacy is protected
US-17	Laboratory	upload test results and reports to a patient's file	the doctor and patient receive them quickly
US-18	Laboratory	access only the data related to my examination	patient privacy is protected
US-19	Administrator	manage user accounts	only valid users use the system
US-20	Administrator	review and verify doctors' medical licenses	only real doctors are accepted
US-21	Administrator	monitor system performance without seeing medical data	the system runs well and patient privacy is kept

2.2 Use Cases

ID	Use Case	Actor	Preconditions	Main Flow
UC-01	Register and Log In	Patient, Doctor, Laboratory, Receptionist (log in only)	User has no account (register) or has an account (login)	User enters data → system validates → account created / user logged in → system opens the dashboard for the user's role
UC-02	Manage Access Permission	Patient	Patient is logged in	Patient opens permissions page → selects a doctor → grants or revokes access → system saves and applies the change
UC-03	Book Appointment	Patient	Patient is logged in	Patient searches doctors → selects a doctor and time → confirms → system saves the appointment
UC-04	View Medical Record	Patient	Patient is logged in	Patient opens medical record → system shows diagnoses, prescriptions, and lab results
UC-05	View Patient History	Doctor	Doctor is verified and the patient granted access	Doctor opens patient list → selects a patient → system checks permission → shows history (read-only)
UC-06	Add Diagnosis and Prescription	Doctor	Doctor has access to the patient	Doctor writes a diagnosis and prescription → saves → system adds them to the patient's record
UC-07	Upload Lab Result	Laboratory	Laboratory is logged in	Laboratory selects the patient → uploads result or report → system attaches it to the patient's file
UC-08	Verify Doctor License	Administrator	A doctor has registered	Admin opens pending requests → reviews the license → approves or rejects → system updates the doctor's status
UC-09	Get Specialty Suggestion	Patient	Patient is logged in	Patient enters symptoms → AI suggests suitable specialties → system lists available doctors
UC-10	Search on Map	Patient	Patient is logged in	Patient opens the map → system shows nearby doctors and laboratories → patient selects one to view details
UC-11	View Doctor Dashboard	Doctor	Doctor is logged in	Doctor opens dashboard → system shows schedule and charts (common diseases, adherence, patient progress)
UC-12	Manage Clinic Team	Doctor	Doctor is logged in and verified	Doctor opens clinic team page → adds a receptionist account → sets its permissions → system saves and links the account to the doctor
UC-13	Manage Patient Appointments	Receptionist	Receptionist is logged in and linked to a doctor	Receptionist selects a patient → chooses a day and time → books, reschedules, or cancels → system saves the change
UC-14	Manage Daily Queue	Receptionist	Receptionist is logged in and the patient has an appointment today	Receptionist opens the queue → checks the patient in → system updates the patient's status (arrived / in progress / done)

ID	Use Case	Actor	Preconditions	Main Flow
UC-15	Request Patient Access	Receptionist, Patient	Receptionist is logged in; patient has the app or a phone	Receptionist asks for the access request → patient shows a QR code or enters an OTP → patient approves → system opens access for the doctor only, without showing medical data to the receptionist

3. Functional Requirements

3.1 Introduction

This document lists the features and functions the system must provide, grouped by module, with a priority for each requirement.

3.2 Requirements

ID	Module	Requirement	Priority
FR-01	Authentication	The system shall allow users to register as a patient, doctor, or laboratory.	High
FR-02	Authentication	The system shall allow users to log in and log out securely.	High
FR-03	Authentication	The system shall allow users to reset a forgotten password.	Medium
FR-04	Access Control	The system shall show pages and actions according to the user's role and block unauthorized access.	High
FR-05	Patient	The patient shall be able to view their own medical record.	High
FR-06	Patient	The patient shall be able to grant or revoke a doctor's access to their record at any time.	High
FR-07	Patient	The patient shall be able to search for doctors by specialty or name.	High
FR-08	Patient	The patient shall be able to book, view, and cancel appointments.	High
FR-09	Doctor	The doctor shall see the list of patients who granted access.	High
FR-10	Doctor	The doctor shall have read-only access to a permitted patient's medical history.	High
FR-11	Doctor	The doctor shall be able to add diagnoses and electronic prescriptions.	High
FR-12	Doctor	The doctor shall be able to view and manage their appointment schedule.	Medium
FR-13	Doctor	The doctor shall be able to add a receptionist account to their clinic and define its permissions.	High
FR-14	Receptionist	The receptionist shall be able to book, reschedule, and cancel appointments for the doctor or clinic they are linked to.	High
FR-15	Receptionist	The receptionist shall be able to view the daily queue and record patient check-in and visit status.	High
FR-16	Receptionist	The receptionist shall be able to request access to a patient's record on behalf of the doctor, and the patient shall approve it using OTP or QR code.	Medium
FR-17	Receptionist	The system shall limit the receptionist to administrative data (name, phone, appointment) and block diagnoses, prescriptions, and results.	High
FR-18	Laboratory	The laboratory shall be able to upload results and reports to a patient's file.	High
FR-19	Laboratory	The system shall prevent the laboratory from editing prescriptions or viewing unrelated medical history.	High
FR-20	Laboratory	The laboratory shall be able to view a log of its uploaded results.	Medium

ID	Module	Requirement	Priority
FR-21	Administrator	The administrator shall be able to create, suspend, and delete accounts.	High
FR-22	Administrator	The administrator shall be able to verify or reject doctors' medical licenses.	High
FR-23	Administrator	The administrator shall be able to monitor system performance without access to medical data.	Medium
FR-24	Map	The system shall show doctors and laboratories on an interactive map.	Medium
FR-25	Dashboard	The doctor dashboard shall show the most common diseases among the doctor's patients this month.	Medium
FR-26	Dashboard	The doctor dashboard shall show the patients' adherence rate to treatment doses.	Medium
FR-27	Dashboard	The doctor dashboard shall show a chart of a patient's condition over time (e.g., blood sugar, blood pressure).	Medium
FR-28	AI Assistant	The system shall let the patient enter symptoms through a chatbot or a form.	Medium
FR-29	AI Assistant	The system shall suggest a suitable specialty and list available doctors, without giving a medical diagnosis.	Medium
FR-30	Security	The system shall store sensitive medical data in encrypted form.	High

4. Non-Functional Requirements

4.1 Introduction

This document defines the quality criteria of the system: security, performance, usability, and reliability.

4.2 Requirements

ID	Category	Requirement
NFR-01	Security	All communication between the browser and the server shall use HTTPS.
NFR-02	Security	Sensitive medical data shall be encrypted at rest.
NFR-03	Security	Passwords shall be stored hashed, never as plain text.
NFR-04	Security	Every request shall be checked against the user's role and the patient's access permission.
NFR-05	Privacy	The administrator shall not be able to view medical data, a laboratory shall only access data related to its own examination, and a receptionist shall only see administrative data (name, phone, appointment).
NFR-06	Performance	Pages shall load in less than 3 seconds under normal use.
NFR-07	Performance	Common operations (login, booking, record view) shall respond in less than 2 seconds.
NFR-08	Usability	The interface shall be simple enough to be used without training.
NFR-09	Usability	The interface shall be responsive and work on mobile, tablet, and desktop screens.
NFR-10	Reliability	The system shall be available during testing and the final demo (target uptime 99%).
NFR-11	Reliability	Database backups shall be taken regularly to prevent data loss.
NFR-12	Compatibility	The system shall work on the latest versions of Chrome, Edge, and Firefox.
NFR-13	Maintainability	The code shall be organized in separate modules and stored on GitHub with version control.
NFR-14	Scalability	The architecture shall allow adding new services without redesigning the whole system.
NFR-15	Safety and Legal	The AI assistant shall only suggest a specialty and shall show a clear notice that it does not provide a diagnosis.